

Assignment -3

Marks - 10

Write a C or C++ program that:

1. Accepts an array of n integers (generated randomly).
 2. Sorts the array using the following algorithms separately:
 - Quick Sort
 - Heap Sort
 - Merge Sort
 3. Measures and displays the following for each algorithm:
 - Execution Time (in milliseconds)
 - Number of Swaps
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Input Format

10
15 698 -32 26985 22 46641 -725 47852 12 5698

Output Format

Quick Sort:
Execution Time: 23 ms
Number of Swaps: 59

Heap Sort:
Execution Time: 20 ms
Number of Swaps: 50

Merge Sort:
Execution Time: 29 ms
Number of Swaps: 70

Note:

The above data is only for illustration purposes and does not represent actual results.
Execution time and swap counts will vary depending on system specifications and input data.

Important Instructions

- The program must be implemented **only in C or C++**.
- Since the assignment involves **execution time measurement**, **portal submission is not permitted**.

- Submit your assignment through the following link:
 <https://forms.gle/FdcFHKCCSneSxEZA>
- **Submission Deadline:** *05-November-2025*
- **Only one submission** per student is allowed.